

Unit 7, Lesson 8: Writing Equations of Linear Relationships

Benchmarks

MA.8.AR.3.2 Given a table, graph, or written description of a linear relationship, determine the slope.

MA.8.AR.3.3 Given a table, graph or written description of a linear relationship, write an equation in slope-intercept form.

Learning Targets

- ☐ I can write the equation of a linear relationship in slope-intercept form given a graph or a table of values.
- ☐ I can determine the slope of a line from a graph or a table, given a mathematical context.

7.8.1 Warm-Up: The Need for Speed

The fastest jet in the world is the Lockheed SR-71 Blackbird. On its last flight, March 6, 1990, a record speed was recorded by flying from Los Angeles, CA to Washington, D.C. in 1 hour, 4 minutes, and 20 seconds.

A typical passenger airplane completed the same route in about 4 hours and 45 minutes.

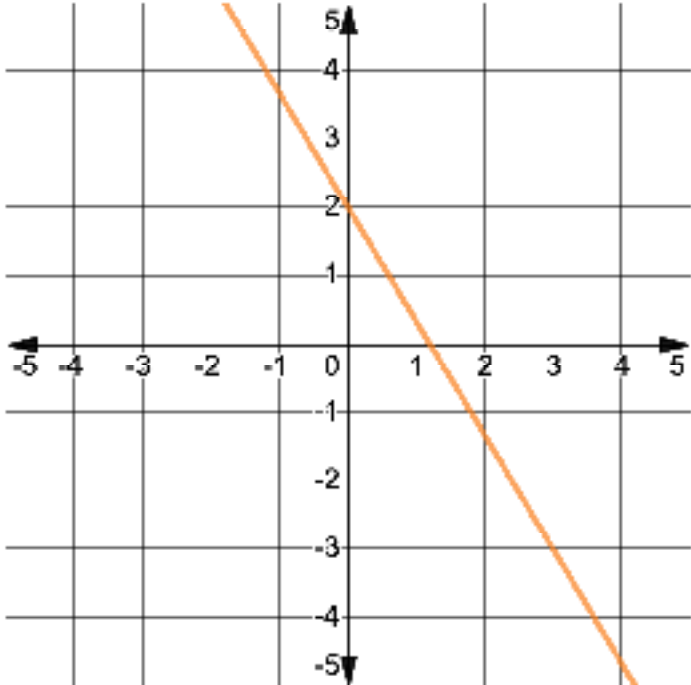
The flight distance between Los Angeles, CA and Washington, D.C. is 2,280 miles.



1. What was the Blackbird's average speed when it set the record speed? Round your answer to the hundredths place.

7.8.2 Guided Practice: Writing the Equation of a Linear Relationship

1. For each linear relationship shown, work with your partner to determine the slope and y-intercept of the line.

Linear Relationship A	Slope	y-intercept
		

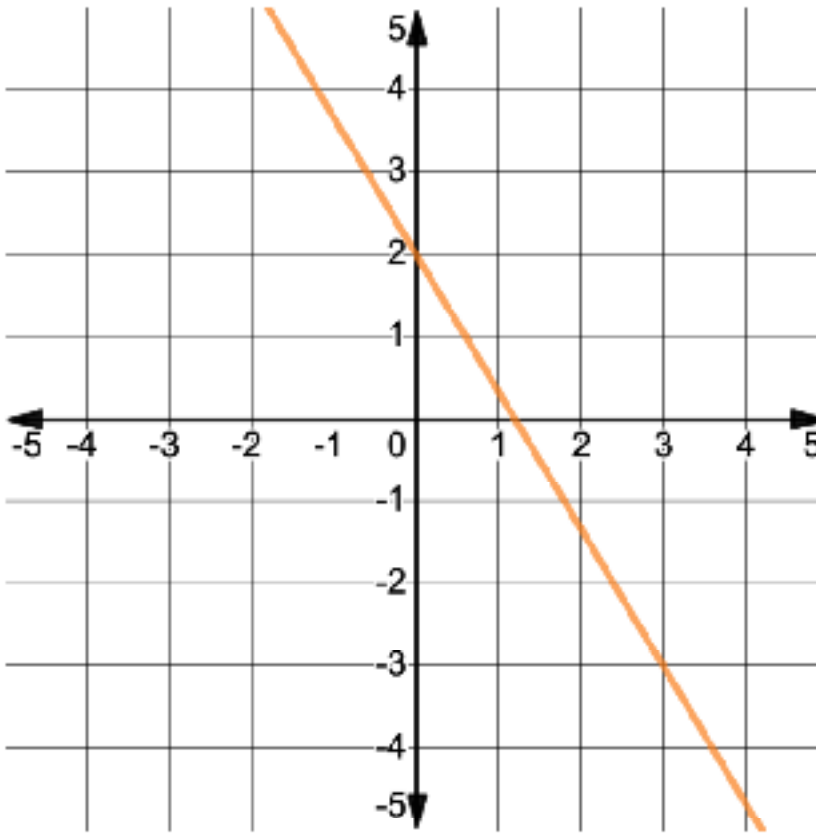
Linear Relationship B	Slope	y-intercept										
<table><tr><td>x</td><td>-10</td><td>0</td><td>3</td><td>4.5</td></tr><tr><td>y</td><td>2</td><td>6</td><td>7.2</td><td>7.8</td></tr></table>	x	-10	0	3	4.5	y	2	6	7.2	7.8		
x	-10	0	3	4.5								
y	2	6	7.2	7.8								

Linear Relationship C	Slope	y-intercept



Slope-intercept form of linear equations is written in the form $y = mx + b$, where m is the slope, or rate of change, of the graph, and b is the y-intercept of the graph.

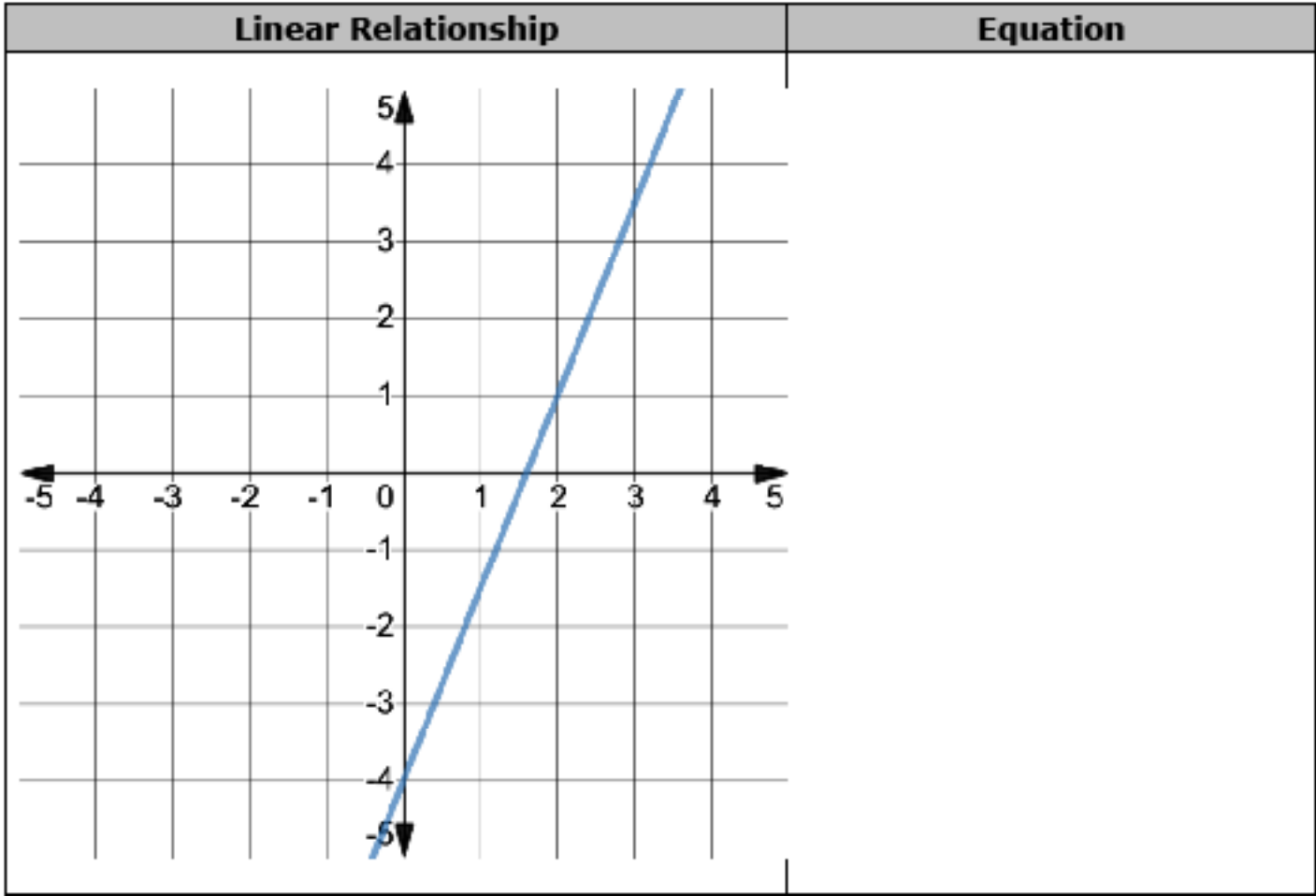
2. Use the slope and y-intercept for each linear relationship to write the equation of each line in slope-intercept form.

Linear Relationship	Equation										
											
<table><tr><td>x</td><td>-10</td><td>0</td><td>3</td><td>4.5</td></tr><tr><td>y</td><td>2</td><td>6</td><td>7.2</td><td>7.8</td></tr></table>	x	-10	0	3	4.5	y	2	6	7.2	7.8	
x	-10	0	3	4.5							
y	2	6	7.2	7.8							

7.8.3 Activity: Gallery Walk

Several students wrote equations to represent linear relationships given as tables or graphs, but each student made at least one error.

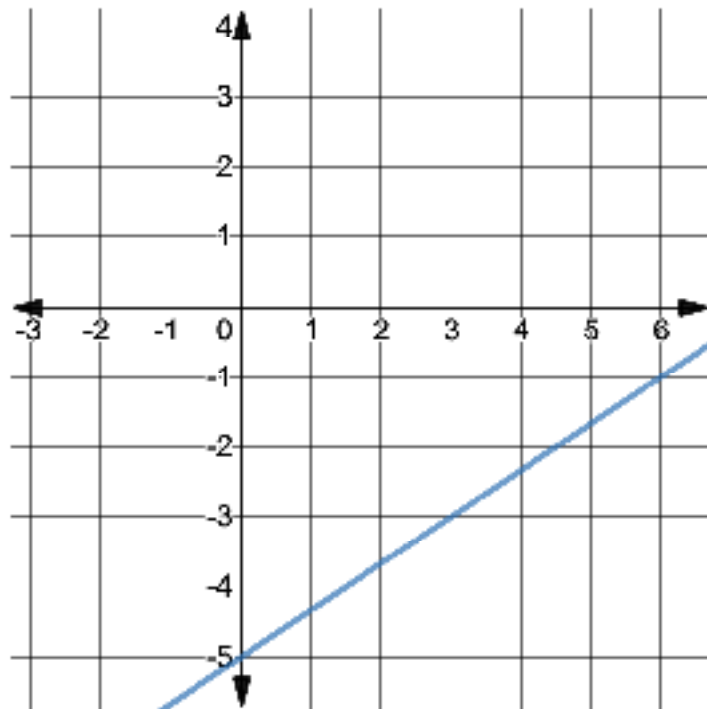
With your partner or small group, complete the table by analyzing each student’s work to identify their error(s). Then write the correct equation for each linear relationship.



Student’s Name	What error(s) did they make?	Correct Equation
Sophia		
Austin		
Caleb		
Hailey		

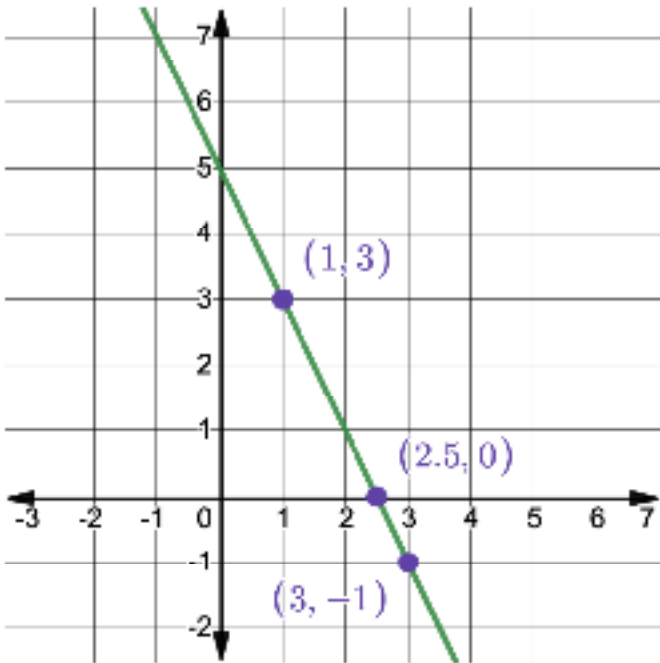
7.8.4 Your Turn!

1. Write the equation of each line in slope-intercept form.

Linear Relationship	Equation										
<table><tr><td><i>x</i></td><td>-4</td><td>-2</td><td>0</td><td>2</td></tr><tr><td><i>y</i></td><td>2</td><td>$\frac{5}{4}$</td><td>$\frac{1}{2}$</td><td>$-\frac{1}{4}$</td></tr></table>	<i>x</i>	-4	-2	0	2	<i>y</i>	2	$\frac{5}{4}$	$\frac{1}{2}$	$-\frac{1}{4}$	
<i>x</i>	-4	-2	0	2							
<i>y</i>	2	$\frac{5}{4}$	$\frac{1}{2}$	$-\frac{1}{4}$							
											

7.8.5 Wrap-Up: Ticket Out the Door

1. The graph and table each represent linear relationships. Write the equation of each line in slope-intercept form.

Linear Relationship	Equation														
															
<table><tr><td><i>x</i></td><td><i>y</i></td></tr><tr><td>-4</td><td>-5</td></tr><tr><td>0</td><td>-3</td></tr><tr><td>1</td><td>-2.5</td></tr><tr><td>3</td><td>-1.5</td></tr><tr><td>6</td><td>0</td></tr><tr><td>10</td><td>2</td></tr></table>	<i>x</i>	<i>y</i>	-4	-5	0	-3	1	-2.5	3	-1.5	6	0	10	2	
<i>x</i>	<i>y</i>														
-4	-5														
0	-3														
1	-2.5														
3	-1.5														
6	0														
10	2														

